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Deep-learning CT Reconstruction in Clinical Scans of the Abdomen: A Systematic Review and Meta-Analysis

Mostafa A Shehata, MD¹, Abdelrahman Mohamed², Serageldin Kamel, MD³, Nir Stanietzky, MD¹, Alicia M Roman-Colon, M.D.⁴, Ajaykumar C Morani, MD¹, Khaled M Elsayes, MD¹, Corey T Jensen, MD¹

¹department of Abdominal Imaging, The University of Texas MD Anderson Cancer Center, 1400 Pressler Street, Unit 1473, Houston, TX 77030-4009

²Alexandria University, Faculty of Medicine, Alexandria, Egypt

³department of Lymphoma and Myeloma, The University of Texas MD Anderson Cancer Center, 1400 Pressler Street, Unit 1473, Houston, TX 77030-4009

⁴Texas Children's Hospital, Houston, TX

Abstract

Objective: To perform a systematic literature review and meta-analysis of the two most common commercially available deep learning algorithms for CT.

Methods: We used PubMed, Scopus, Embase, and Web of Science to conduct systematic searches for studies assessing the most common commercially available deep learning CT reconstruction algorithms: True Fidelity (TF) and Advanced intelligent Clear-IQ Engine (AiCE) in the abdomen of human participants since only these two algorithms currently have adequate published data for robust systematic analysis.

Results: Forty-four articles fulfilled inclusion criteria. 32 studies evaluated TF and 12 studies assessed AiCE. DLR algorithms produced images with significantly less noise (22-57.3% less than IR) but preserved a desirable noise texture with increased contrast-to-noise ratios and improved lesion detectability on conventional CT. These improvements with DLR were similarly noted in dual-energy CT which was only assessed for a single vendor. Reported radiation reduction potential was 35.1-78.5%. Nine studies assessed observer performance with the two dedicated liver lesion studies being performed on the same vendor reconstruction (TF). These two studies indicate preserved low contrast liver lesion detection (>5mm) at CTDI_{vol} 6.8 mGy (BMI 23.5 kg/m²) to 12.2 mGy (BMI 29 kg/m²). If smaller lesion detection and improved lesion characterization is needed, a CTDI_{vol} of 13.6 mGy – 34.9 mGy is needed in a normal weight to obese population. Mild signal loss and blurring have been reported at high DLR reconstruction strengths.

Address correspondence to C.T.J., cjensen@mdanderson.org, 713 745-1152 (f).

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Conclusion: Deep learning reconstructions significantly improve image quality in CT of the abdomen. Assessment of other dose levels and clinical indications is necessary. Careful choice of radiation dose levels is necessary, particularly for small liver lesion assessment.

Keywords

CT; Image reconstruction; abdomen; deep learning; artificial intelligence; dose reduction; convolutional neural network

Background

Deep learning reconstructions (DLR) are the most recent development in CT reconstruction technology. DLRs are increasingly replacing iterative reconstructions (IR) which have provided dramatic noise magnitude reduction compared to filtered back projection (FBP) but IRs are associated with an undesirable noise texture, particularly at low radiation doses [1, 2]. In the setting of radiation dose reduction, low contrast tasks have been constrained with prior studies demonstrating less dose reduction potential than initially estimated for IR [1, 3, 4]. Recent evidence suggests significant image quality improvement using DLR [2, 5-7].

Over a relatively short time frame, a large body of evidence has formed showing the robust denoising of DLRs [8]. Multiple phantom analyses have been performed on both Advanced Intelligent Clear-IQ Engine (AiCE) and TrueFidelity (TF). These studies have also indicated a strong reduction in noise magnitude, improved lesion detectability and maintained high-contrast spatial resolution. The algorithms preserve image noise texture with TF targeting the appearance of FBP and AiCE targeting model-based iterative reconstruction based on their respective data training sets [9-14]. Mixed results have been reported regarding low contrast spatial resolution at reduced doses [15]. For example, Racine et al. reported preserved low contrast liver lesions at lower doses in an anthropomorphic phantom with model observer using TF indicating a dose reduction potential of 30% compared to ASIR-V [16]. Other phantom analyses have indicated a higher dose reduction potential of 50 - 70% for TF [11, 14, 17] and 52% for AiCE compared to hybrid IR [18]. However, Solomon et al. identified decreased low contrast spatial resolution using a custom rod phantom in TF compared to FBP, albeit still improved compared to ASIR-V [9]. Similar findings have been seen with AiCE. In a comparative phantom analysis of TF and AiCE, Greffier et al. showed an undesirable smoothing when using the strong level of AiCE that was even more pronounced at lower radiation doses [10]. While Greffier et al. showed minimal effect of dose and contrast on spatial resolution with TF, the authors conversely showed that AiCE was non-linear, similar to IRs. The authors suggest this reconstruction behavior likely stems from their respective underlying data set training; TF is a direct DLR algorithm trained with FBP scans while AiCE is an indirect algorithm trained with studies using non-linear model-based IR [2]. Overall, the authors showed TF to have better image quality with superior lesion detectability compared to AiCE [10].

While the literature indicates overall improved image quality with DLR, typical of a nascent CT reconstruction, many reports have focused on important but simple initial assessments

such as noise reduction magnitude and contrast-to-noise ratio (CNR). Importantly, the literature has shown reader preference for the resultant noise texture of DLR which is in stark contrast to reports on IR [19, 20]. Mounting evidence for specific clinical tasks also suggests that DLR represents a significant improvement, either allowing further radiation dose reduction or improved image quality at maintained doses [5]. Prospective lesion performance studies are still few [19-21]. Further study of subtle clinical tasks such as low contrast liver lesion detection is needed; Jensen et al. reported that IR could not maintain observer performance for lesions $\leq 10\text{mm}$ at a 54% dose reduction and observer performance was inferior with DLR for lesions $\leq 5\text{mm}$ at 65% dose reduction [1, 19].

There has been a substantial increase in publications since the most recent systematic review of abdominal CT using DLR [22]. Herein, we systematically review the literature for abdominal CT using either of the two most common commercially-available DLR algorithms in human subjects. Despite the fact that there are many deep learning algorithms e.g., TrueFidelity (GE Healthcare), AiCE (Canon Medical Systems), Precise Image (Philips Healthcare), PixelShine (AlgoMedica), ClariCT.AI (ClariPi Inc.) [2], our aim was to provide a meaningful summary of evidence which requires robust data currently only available in significant volume for TF and AiCE.

Methods

This systematic review followed the preferred reporting items for systematic reviews and meta-analysis extension for scoping reviews (PRISMA 2020) guidelines [23].

Literature Search

We conducted bibliographic searches in PubMed, Scopus, Embase, and Web of Science on the 23rd of January, 2023. No beginning search date was set. The following search string was used: ("deep Learning" OR "neural network" OR "autoencoder" OR "CNN" OR "artificial intelligence") AND ("reconstruction" OR "denoising" OR "reformation") AND ("CT" OR "Computed Tomography") AND ("abdomen" OR "abdominal" OR "liver" OR hepat* OR "kidney" OR "renal" OR "pancreas" OR "adrenal" OR "sple*"). References cited in review articles were also examined manually for related articles, and the duplicates were removed.

Eligibility Criteria and Data Extraction

The titles and abstracts of all studies were independently screened by two researchers followed by a second screening using the full article text. If the researchers did not agree, a final decision was made by a third researcher.

Studies were included if they met the following criteria: 1) full-text original article written in English; 2) assessed one of the two major commercially available DLR algorithms: True Fidelity (TF, GE Healthcare) or Advanced Intelligent Clear-IQ Engine (AiCE, Canon Medical Systems) to other reconstruction methods; 3) used CT as an imaging modality for the abdominal organs; 4) assessed human subjects.

Studies were excluded if they met the following criteria: 1) deep learning reconstruction algorithms other than TF or AiCE; while many other deep learning algorithms exist [2], our

aim was to provide a meaningful summary of evidence which requires robust data currently only available in significant volume for TF and AiCE 2) studies using open-access or public datasets; 3) studies using non-CT imaging modalities; 4) positron emission tomography CT and cone beam CT; 5) case reports, reviews, conference abstracts, poster presentations, and opinion papers; 6) phantom-only studies.

Data Extraction and Analysis

Articles meeting eligibility criteria were reviewed by three authors. Key characteristics such as study type (e.g., prospective, retrospective), human subjects' age, scan type (e.g., contrast-enhanced, dual energy), anatomy evaluated, reconstruction algorithms used, study radiation doses, kVp, and body mass index (BMI) for the included studies were summarized and used for categorization. Additional reported characteristics include: last name of the first author and year of study publication, title, study type (prospective or retrospective), number of patients, and anatomy evaluated. For outcomes, we extracted the study aim, radiation dose reduction, quantitative outcome measures, qualitative outcome measures, number of readers, key results, conclusions, noise reduction, and limitations.

Quality Assessment

Each included study was evaluated using a custom-made quality assessment made by Stiphout and colleagues (Appendix) [22]. Studies were assessed for study design, data collection, study sample, statistical analysis, funding, scan settings, type of reconstruction comparison and region of interest description. A list of all questions is listed in the Appendix. A point for each question could result in a maximum score of 18. Studies with a score of 6 or lower were classified as low-quality, 7–11 as moderate-quality studies, and 12 or more as high-quality studies.

Statistical analysis

Meta-analyses were performed with MedCalc® Statistical Software version 20.218 using random effects [24]. I^2 statistics were calculated and high heterogeneity was defined as >75% with statistical significance defined as $p < 0.05$. Results were summarized in forest plots.

Results

Literature Search

Our systematic search of the literature identified 1184 articles from the four databases. After removing 549 duplicates and the screening of the remaining 635 titles and abstracts we had 147 provisionally eligible articles. We screened the full text for the remaining articles and yielded 50 potentially eligible articles. Six articles were excluded during extraction as they did not report our outcomes of interest and did not differentiate the abdominal organs from other body organs results. No additional article was identified after searching the bibliographies. Finally, 44 articles fulfilled the selection criteria and were included in the present systematic review (Figure 1).

Baseline Characteristics of the Included Studies

Table 1 and Table 2 show the detailed characteristics of the forty-four included studies. All articles were published in or after 2019 with a mean of 59 ± 31 patients and a mean of 2.6 ± 1.5 readers. Only five studies assessed DECT and all were with TF. There were 27 retrospective studies and 17 prospective studies. 4 studies were on pediatric participants. 32 studies evaluated TF and 12 studies assessed AiCE. For the SECT studies, two were at 80 kVp, six at 100 kVp, twenty-one at 120 kVp, two had unreported kVp, and eight had variable kVp. The mean radiation dose level assessed was $10.5 \text{ mGy} \pm 7.3$. There were 19 studies that compared between a standard dose and reduced dose with a mean reduced dose of $4.7 \text{ mGy} \pm 3.8$ resulting in a mean dose reduction of $62\% \pm 16$.

Outcomes and Quality Assessment

We summarized the results of each study in table 3 and table 4 showing any outcomes measured, conclusion, noise reduction, limitations, and quality assessment scores. The meta-analyses for AiCE and TF image noise reduction compared to iterative reconstructions are shown in figures 2 and 3 with overall mean noise reductions of -2.8 ± 0.5 (95% CI: $-3.7, 1.8$; $P < 0.001$) and -3.2 ± 0.3 (95% CI: $-3.9, 2.6$; $P < 0.001$), respectively. Compared to IR, AiCE reduced mean image noise by $39\% \pm 9$ (range, 26.7-49%) and TF reduced mean image noise by $39\% \pm 13$ (range, 22-57%). For AiCE and TF, radiation reduction potential was reported to be 44.3-76.6% and 35.1-78.5%, respectively. There were eight studies assessing dose reduction using AiCE and eleven using TF with mean radiation dose reduction of $57.6\% \pm 19$ and $65.7\% \pm 14.3$, respectively [18-21, 25-33]. 9 studies (3 AiCE, 6 TF) assessed observer performance for heterogeneous conditions [19-21, 30, 32, 34-37], with only two studies prospectively and specifically assessing liver lesion detection observer performance, both of which used TF [19, 21].

Discussion

Our systematic review and meta-analysis of the two most common and commercially-available DLRs summarized data demonstrating the robust noise reduction capability of both algorithms. Using IRs as the reference standard, AiCE and TF reduced image noise by up to 49% and 57%, respectively.

While denoising is desirable to enhance perceived image quality, diagnostic confidence, and often observer performance, this has historically come with the tradeoff of undesirable altered image noise texture with IR [1]. Our systematic review reveals that readers overwhelmingly prefer DLRs over IRs and FBP at the same dose level. Multiple studies have shown higher image quality ratings even for RD DLR compared to SD IR and FBP [19, 26, 28]. Studies assessing dose reduction in AiCE and TF indicate a mean dose reduction potential of 57.6% and 65.7%, respectively, which correlates with prior phantom analyses.

The benefits of DLRs at SECT extend in a similar fashion to DECT. At present, of the two DLRs, only TF has been assessed using DECT in abdominal CT. Greffier et al. demonstrated the highest detectability (high contrast) with TF on DECT (40 – 80 keV) compared to AV50 and FBP and Fukutomi et al. reported decreased variability of iodine concentration

when using TF compared to AV 30/50 and FBP [38, 39]. The enhanced noise reduction by DLR has specific implications for DECT workflows and capabilities. Where previous noise limitations may have resulted in choices of higher noise keV selections, DLR now allows similar noise but at lower keVs which should improve many diagnoses through increased CNR [30, 31, 40-42].

At high contrast, compared to FBP, Masuda et al. showed a rightward shift of spatial frequencies ($f_{50\%}$) for TF on both SECT (120 kVp) and with DECT 70 keV, more than that seen with hybrid IR [17]. Conversely, at medium contrast, they showed a decreased $f_{50\%}$ of TF at 70 keV, but less than that seen with hybrid IR; there was only a mild leftward shift of $f_{50\%}$ for TF on SECT, much less than with hybrid IR. In concordance with prior reports of mild progressive blurring with increasing TF strengths, Masuda et al. also showed mild decreased $f_{50\%}$ with increasing TF strength. Similar to Jensen et al., the authors suggested choosing TF strength based on the degree of contrast suspected for the primary clinical task [6, 43, 44].

There were 9 studies that assessed observer performance for various conditions (3 AiCE, 6 TF): two assessed participants with hepatic metastases, one assessed pancreatic cancer resectability, four assessed a mixture of abdominal 'lesions', and two assessed urolithiasis. Only 2 studies were prospective with the specific aim of determining liver lesion observer performance [19, 21]. Lyu et al. showed that TF enabled a 50% dose reduction ($CTDI_{vol}$ 6.8 mGy) compared to full-dose FBP and IR. Jensen et al. reported preserved lesion detection for lesions larger than 5mm at 65% dose reduction ($CTDI_{vol}$ 12.2 mGy), but inferior RD TF accuracy. Based on these two studies, for the clinical task of small (>5mm) low contrast liver lesion detection, the likely mean radiation dose needed is between the reduced doses of these two studies ($CTDI_{vol}$ 6.8 mGy at a mean BMI of 23.5 kg/m² – 12.2 mGy at a mean BMI of 29 kg/m²). If sub 5mm lesion detection and improved lesion characterization is needed, the range is likely between the standard doses of these studies ($CTDI_{vol}$ 13.6 – 34.9 mGy). Prospective evaluation by Singh et al. using AiCE demonstrated preserved detection at an 84% dose reduction (standard dose mean $CTDI_{vol}$ of 13 mGy), however, this study was limited by its use of a heterogeneous set of lesions in the abdomen such as including lymph nodes [20]. Compared to the ACR dose index registry, Ahmad et al. recently reported higher CT radiation dose levels in a tertiary oncology-specific center and reported lower image noise than in a practice with a mixture of general and oncology cases [45, 46]. As also discussed by Szczykutowicz and colleagues, it was posited that these differences relate to increased imaging demands in oncology to visualize small low contrast lesions in the abdomen, but further work is needed [47].

While the focus is often on liver evaluation in abdominal imaging due to the relatively higher prevalence of clinically significant lesions in the liver, it should be noted that many studies have also shown significantly improved subjective and quantitative image quality by DLRs for the spleen, kidneys, musculature and vessels. Furthermore, prospective evaluations of urolithiasis by Zhang et al. indicated maintained observer performance at 57% with TF and 77% with AiCE [36, 37]. Interestingly, and supporting other reports of mild signal loss at higher DLR strengths for TF and AiCE, the urolithiasis studies reported a few examples of tiny calculi (< 3mm) which were not seen on reduced dose DLR [36, 37, 48].

The benefits of DLRs were also seen with thin section imaging demonstrating a similar degree of denoising on both standard ($\geq 2.5\text{mm}$) and thin ($0.5\text{--}1.25\text{mm}$) slice reconstructions [49, 50]. For example, Cao et al. 2022 demonstrated noise on TF_{high} at 1.25 mm approaching that of AV50 at 5mm and the TF reconstruction received significantly higher qualitative reader scores [25]. This enhanced thin section capability can be particularly useful such as with challenging vascular evaluations [51].

Obese patients represent another challenging group for diagnostic imaging. Two studies have been specifically performed on obese participants, one using TF on DECT [40] and the other using AiCE on SECT [52], both of which were retrospective demonstrating similar noise and qualitative reader improvements to non-obese studies. Akagi et al. noted lower noise and higher CNR using AiCE in obese patients as compared to non-obese patients using hybrid IR. Fair et al. also specifically discuss how their demonstrated improvements with TF may help the high-risk obese population by expanding imaging capabilities and quality in this group. For example, the aim of their study directly assesses DECT which has been limited by large patient body habitus and the authors indicate that these limitations are mitigated by DLR.

Our systematic review had several limitations that should be noted. First, we limited the scope of this review to the evaluation of the two most common commercially-available DLRs in abdominal CT to obtain enough data for in-depth review, therefore these results do not apply to other algorithms. Second, our review of only abdominal CT scans excludes reports on DLR in other body regions and applications. Third, heterogeneity is present in the included studies because of differences in radiation doses, slice thickness, patient demographics, and contrast usage. Lastly, it should be noted that the majority of present studies have been performed using True Fidelity and no prospective dedicated liver lesion studies or DECT have been performed using AiCE.

Conclusion

The two most common commercially-available DLRs markedly reduce image noise and preserve desirable image texture. A majority of evidence suggests a significant capacity for radiation dose reduction in SECT (TF and AiCE) and DECT (only evaluated with TF). DLRs allow for lower keV usage by mitigating the typically associated increase in image noise. Limited evidence still suggests caution in the detection of small low contrast liver lesions. Although of doubtful significance clinically, tiny urolithiasis can be obscured with DLRs. Next-generation DLR algorithms should address the mild loss of high spatial frequency information in clinical images. More studies are needed to determine the proper dose levels for various clinical tasks.

Registry and Funding:

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Appendix

Methods

Quality Assessment (1 point for each)

1. The study was conducted prospectively
2. The data was collected first-hand
3.
 - a) Number of patients was described
 - b) Anatomical location where data was collected was described
 - c) Sample size > 10
 - d) Sample size > 40
 - e) Sample size > 60
 - f) Sample size > 100
4.
 - a) Statistical analysis was performed
 - b) Multiple test correction was done
 - c) Results contained descriptive statistic results
5. Funding was described
6.
 - a) CT scanner type was described
 - b) CT scan protocol was described
7.
 - a) DLR was compared with IR
 - b) DLR was compared with FBP
8.
 - a) Images were linked for ROI placement
 - b) Size of ROIs listed

Results

	Cao et al. 2020 [15]	Cao et al. 2022 [51]	Delabie et al. 2022 [46]	Fair et al. 2022 [36]	Fukutomi et al. 2022 [35]	Ichikawa et al. 2021 [52]	Jensen et al. 2019 [6]	Jensen et al. 2022 [9]	Kaga et al. 2021 [53]	Kaga et al. 2022 [16]	Li et al. 2021 [17]	Lyu et al. 2021 [24]	Lyu et al. 2022 [11]
1	1	0	0	0	1	0	0	1	1	1	1	0	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1
3a.	1	1	1	1	1	1	1	1	1	1	1	1	1
3b.	1	1	1	1	1	1	1	1	1	1	1	1	0
3c.	1	1	1	1	1	1	1	1	1	1	1	1	1
3d.	0	0	1	1	1	1	1	1	1	1	1	1	1
3e.	0	0	1	0	0	0	0	0	0	0	0	0	1

3f.	0	0	0	0	0	0	0	0	0	0	0	0	1
4a.	1	1	1	1	1	1	1	1	1	1	1	1	1
4b.	0	0	0	0	1	0	0	0	0	1	0	1	0
4c.	1	1	1	1	1	1	1	1	1	1	1	1	1
5	0	1	1	0	1	0	0	1	0	0	0	1	1
6a.	1	1	1	1	1	1	1	1	1	1	1	1	1
6b.	1	1	1	1	1	1	1	1	1	1	1	1	1
7a.	1	1	1	1	1	1	1	1	1	1	1	1	1
7b.	0	0	1	0	1	0	0	1	0	0	1	1	1
8a.	1	1	0	0	0	0	1	0	0	0	0	0	0
8b.	1	1	1	0	0	1	1	0	0	0	1	0	0
Total	12	12	14	10	14	11	12	13	11	12	13	13	14
	Sato et al. 2022 [60]	Son et al. 2022 [61]	Wang et al. 2021 [62]	Wang et al. 2022 [63]	Xie et al. 2021 [64]	Xu et al. 2022 [38]	Yang et al. 2021 [25]	Yoo et al.2022 [65]	Yoon et al. 2021 [66]	Zhang et al. 2022 [67]	Akagi et al. 2019 [68]	Akagi et al. 2020 [50]	Brady et al. 2021 [14]
1	0	0	1	0	0	0	0	0	0	1	0	0	0
2	1	1	1	1	1	1	1	1	1	1	1	1	1
3a.	1	1	1	1	1	1	1	1	1	1	1	1	1
3b.	1	1	1	1	1	1	1	1	1	1	1	1	1
3c.	1	1	1	1	1	1	1	1	1	1	1	1	1
3d.	0	1	1	1	0	0	1	1	0	0	1	1	0
3e.	0	0	1	1	0	0	0	1	0	0	0	0	0
3f.	0	0	0	0	0	0	0	1	0	0	0	0	0
4a.	1	1	1	1	1	1	1	1	1	1	1	1	1
4b.	0	1	1	1	1	0	1	1	1	1	1	0	0
4c.	0	1	1	1	1	1	0	1	1	1	1	1	1
5	1	1	1	1	1	1	1	0	1	1	1	1	0
6a.	1	1	1	1	1	1	1	1	1	1	1	1	1
6b.	1	1	1	1	1	1	1	1	1	1	1	1	0
7a.	1	1	1	1	1	1	1	1	1	1	1	1	1
7b.	0	1	1	0	0	0	0	1	0	0	0	0	1
8a.	1	1	0	0	1	1	1	1	1	1	0	0	1
8b.	1	1	0	1	0	1	1	0	1	1	1	0	0
Total	11	15	15	14	12	12	13	15	13	14	13	11	10

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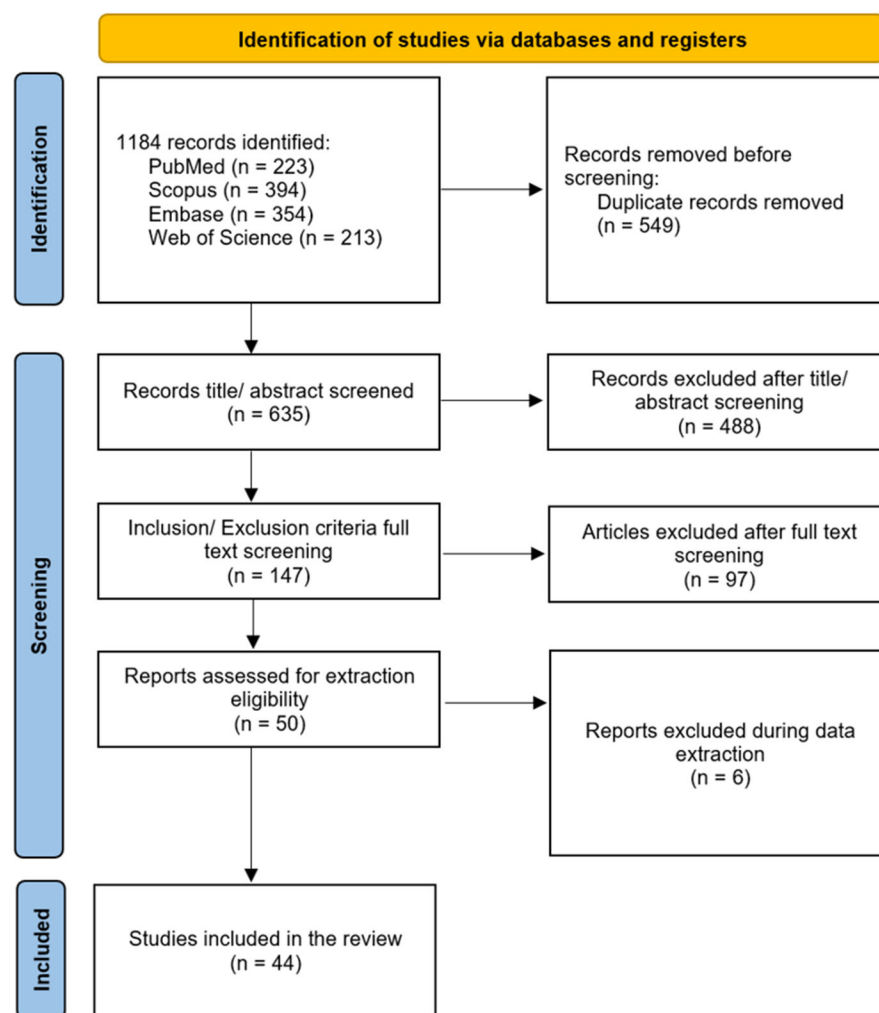


Figure 1.
PRISMA flow chart of the search and screening process

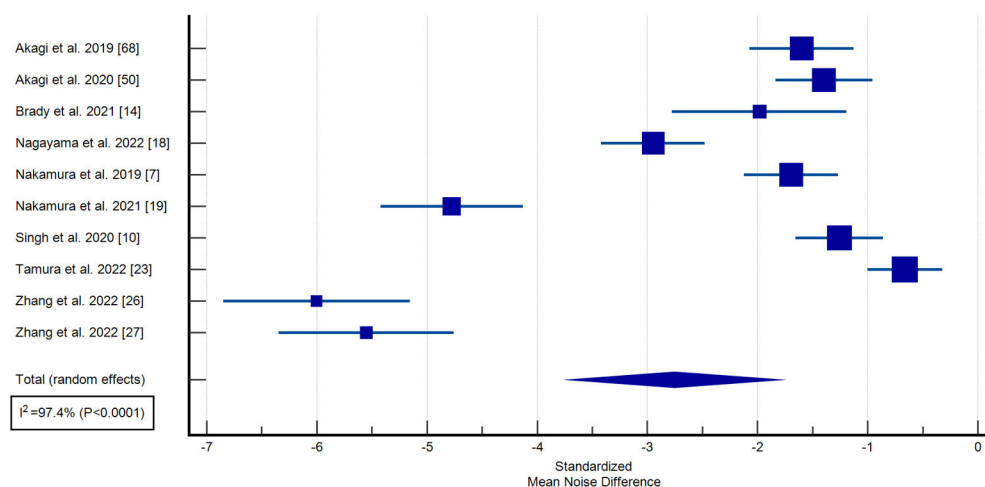


Figure 2.

Forest plot of the standardized mean noise difference between deep learning reconstruction (AiCE) and iterative reconstruction (AiDR) compared at the same dose level.

Note- Tamura et al. 2021 was excluded from meta-analysis since AIDR was not assessed and Oostveen et al. 2022 was excluded due to lack of SD/noise range reporting (only confidence interval available)

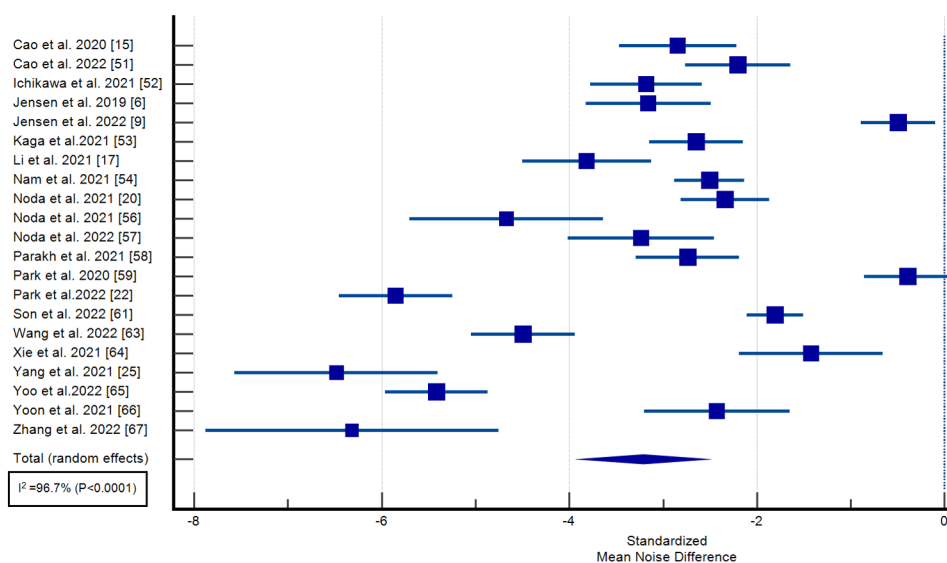


Figure 3.

Forest plot of the standardized mean noise difference between deep learning reconstruction (True Fidelity) and iterative reconstruction (ASIR-V) compared at the same dose level.

Note- Delabie et al. 2022, Fair et al. 2022, Fukutomi et al. 2022, Kaga et al. 2022, Lyu et al. 2021, Lyu et al. 2022, Njølstad et al. 2021, Sato et al. 2022, Wang et al. 2021, and Xu et al. 2022 were excluded from meta-analysis due to incomplete noise data.

Table 1.

Characteristics for studies assessing TrueFidelity (TF) in abdominal CT

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
Cao et al. 2020 [25]	DLIR to improve image quality of extremely low-dose CT for hepatic lesions	Prospective	Adult	Hepatic lesions	40	CECT	Liver Spleen Muscle	High	ASIR-V 50%	Arterial Phase = 6.8 mGy Delayed Phase = 1.7 mGy	120	66 ± 10 kg (Range, 45–85 kg) No BMI provided
Cao et al. 2022 [53]	Improving spatial resolution and diagnostic confidence with thinner slice and DLIR	Retrospective	Adult	Hepatic lesions	40	CECT	Liver	Medium High	ASIR-V 50%	Not reported	120	66 ± 10 kg (Range, 45–85 kg) No BMI provided
Delabie et al. 2022 [48]	Diagnostic performance and image quality of DLIR on low-dose for urolithiasis	Retrospective	Adult	Suspected/known urolithiasis	75	NCCT	Urinary tract	High	ASIR-V 70% FBP	Mean = 2.6 mGy	100	27.2 ± 5.6 kg/m ² (Range=17.9–46.3 kg/m ²)
Fair et al. 2022 [40]	Image Quality Evaluation in DECT in Obese Patients with DLIR	Retrospective	Adult	Obese, oncologic staging	48	CE - DECT	Liver Aorta	Medium High	ASIR-V (unspecified %)	Not reported	DECT 80 140	39.5 ± 7.36 kg/m ²
Fukutomi et al. 2022 [39]	DLIR to improve accuracy of iodine quantification and image quality in DECT	Prospective	Adult	Suspected liver or pancreas lesions	50	CE - DECT	Liver Pancreas Aorta Portal vein	Medium	ASIR-V 50% and 30% FBP	Mean = 10.8 ± 2.4 mGy (Range, 9.2–17.6 mGy)	DECT 80 140	22.5 ± 3.2 kg/m ² ; (Range, 16.8–34.7 kg/m ²)
Ichikawa et al. 2021 [54]	DLIR for improvement of image quality: comparison with ASIR-V	Retrospective	Adult	Oncologic staging or screening	50	CECT	Liver Portal vein Aorta	High	ASIR-V 50%	Median = 15.1 mGy (IQR, 11.9–20.9 mGy)	120	23.0 ± 4.9 kg/m ² (Range, 15.9–33.0 kg/m ²)
Jensen et al. 2019 [6]	Image Quality Assessment by Use of DLIR: Initial Experience	Retrospective	Adult	Oncologic staging	40	CECT	Liver Aorta Spleen	Low Medium High	ASIR-V 30%	Mean = 28.3 mGy (Range, 20.0–49.1 mGy)	120	27 ± 5 kg/m ² (Range, 20–41 kg/m ²)

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
Jensen et al. 2022 [19]	Reduced-Dose DLIR for Liver Metastases	Prospective	Adult	Colorectal cancer liver metastases	51	CECT	Liver	Medium	ASIR-V 50% FBP	Mean= 34.9 mGy ± 10.9 (Range, 10.1–54.9 mGy)	120	29 kg/m ² ± 6 (Range, 18–43 kg/m ²)
Kaga et al. 2021 [55]	DLIR in dynamic contrast-enhanced abdominal CT: image quality and lesion and lesion detection	Prospective	Adult	Screening for liver disease and/or lesions	59	CECT	Liver	Low Medium High	ASIR-V 40%	Not reported	120	59.3 ± 13.2 kg (Range, 39–108 kg) No BMI provided
Kaga et al. 2022 [26]	Unenhanced low-dose CT with DLIR: image quality and anatomical structure depiction	Prospective	Adult	Unspecified	58	NCCT	Liver Pancreas Spleen Kidney	High	ASIR-V 40%	SDCT group mean = 14.8 mGy (Range, 11.3 – 17.9 mGy) LDCT group mean = 3.1 mGy (Range, 2.8–3.6 mGy)	120	SDCT 22.0 ± 0.5 kg/m ² (15.8–29.7 kg/m ²) LDCT 22.2 ± 0.5 kg/m ² (12.5–28.1 kg/m ²)
Li et al. 2021 [27]	A feasibility study of realizing low-dose using DLIR	Prospective	Adult	Unspecified	47	CECT	Spleen Kidney Paraspinal muscles Liver Aorta	Medium High	ASIR-V 80% and 40% FBP	Early arterial phase mean = 20.3 mGy ± 0.9 Late arterial phase mean = 8.6 mGy ± 3.3	80 for < 24 BMI; 100 for > 24	23.2±3.2 kg/m ² (range, 16.02–29.40 kg/m ²)
Lyu et al. 2021 [34]	Effect of DLIR in the prediction of resectability of pancreatic cancer: Diagnostic	Retrospective	Adult	Pancreatic adenocarcinoma	47	CECT	Pancreas	Low Medium	ASIR-V 60% FBP	Mean = 12.0 mGy ± 4.1	120	28.1 ± 6.2 kg/m ²

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
	performance and reader confidence											
Lyu et al. 2022 [21]	Low-dose DLIR for detection of liver metastases on CT routinely?	Prospective	Adult	Hepatic metastases and benign lesions	154	CECT	Liver	Medium	ASIR-V 60% FBP	N/A	120	23.5 ± 4.0 kg/m ²
Nam et al. 2021 [56]	DLIR of the upper abdomen: similar image quality with lower radiation dose in direct comparison with iterative reconstruction	Retrospective	Adult	Oncologic staging or surveillance	100	CECT	Liver Lung Pancreas Biliary	High	ASIR-V 40% and ASIR-V 50%	mean = 7.1 mGy ± 1.9	100	Not provided
Njølstad et al. 2021 [57]	Improved image quality with DLIR - initial clinical experience	Retrospective	Adult	Routine follow-up of underlying malignant condition	10	CECT	Liver	Medium High	ASIR-V 50%	Mean = 14.7 mGy (Range, 8.3 – 27.0 mGy)	120	Not provided
Noda et al. 2021 [30]	Low-dose whole-body CT using DLIR: image quality and lesion detection	Prospective	Adult	Staging/restaging	59	CECT	Liver Pancreas Spleen Aorta	High	ASIR-V 40%	Low dose group mean = 2.9 mGy ± 0.8 Standard dose mean = 13.5 mGy ± 4.1	120	20.8 ± 3.4 kg/m ²
Noda et al. 2022 [58]	DLIR for pancreatic low-dose computed tomography: comparison with ASIR-V	Prospective	Adult	Pancreatic ductal adenocarcinoma	28	CECT	Pancreas	Medium High	ASIR-V 40%	Mean = 2.3 mGy ± 1.0 (Range, 1.1–5.0 mGy)	120	20.3 kg/m ² (Range, 15.1–32.0 kg/m ²)
Noda et al. 2022 [59]	DLIR for pancreatic protocol DECT: image quality and quantification of iodine concentration	Prospective	Adult	Pancreatic ductal adenocarcinoma, two patients with pancreatic anaplastic carcinoma	30	CE - DECT	Pancreas Liver Abdominal aorta Aortal vein	Medium High	ASIR-V 40%	Not reported	DECT 80 140	22 ± 4 kg/m ² (Range, 16–32 kg/m ²)

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
Noda et al. 2022 [31]	Radiation and iodine dose reduced DECT at 40 keV reconstructed with DLIR	Prospective	Adult	Staging/restaging	111	CE – SECT and DECT	Liver Pancreas Spleen Kidney Abdominal aorta Portal vein Hepatic vein Inferior vena cava	High	ASIR-V 40%	SD median = 10 mGy RD median = 4 mGy	120 and DECT 80 140	Standard Dose 22 kg/m ² (Range, 20–23 kg/m ²) Double Low-dose 21 kg/m ² (19–23 kg/m ²)
Parakh et al. 2021 [60]	Sinogram-based DLIR: image quality considerations	Retrospective	Adult	Oncological evaluation (N = 27) Abdominal pain (N = 23)	50	CECT	Liver Portal vein Subcutaneous fat	Low Medium High	ASIR-V 40%	Mean = 10.5 mGy ± 5.4 (Range, 4.5 – 30.3 mGy)	100,120	80.9 ± 21.0 kg No BMI provided
Park et al. 2020 [61]	CT iterative vs DLIR: comparison of noise and sharpness	Retrospective	Adult	Not specified	37	CECT	Liver Aorta Popliteal artery Femoral artery	Low Medium High	ASIR-V 80% and 100%	Mean = 1.4 mGy ± 0.3	70	23.2 kg/m ² , (Range 14.3–32.9 kg/m ²)
Park et al. 2022 [32]	Image Quality and Lesion Detectability of Lower-Dose Obtained Using DLIR	Retrospective	Adult	Restaging	111	CECT	Liver Paraspinal muscle Aorta	Medium High	ASIR-V 40%	Standard dose mean = 7.2 mGy ± 1.7 Low dose mean = 4.6 mGy ± 1.2	100	Standard Dose 63 ± 10 kg (Range, 40–89 kg) Low Dose 63 ± 11 kg (Range, 37–93 kg) No BMI provided
Sato et al. 2022 [62]	DLIR for improving image quality of DECT	Retrospective	Adult	Not specified	40	CE-DECT	Liver	Medium High	ASIR-V 50%	Median = 10.9 mGy (IQR 8.4 – 14.8 mGy)	DECT 80 140	22.6 ± 11.4 kg/m ²
Son et al. 2022 [63]	Comparison DLIR with FBP and ASIR-V for Pediatric Abdominopelvic CT	Retrospective	Pediatric	Not specified	120	CECT	Liver	Low Medium High	ASIR-V 50% and 100% FBP	Mean = 2.9 mGy ± 1.6	Auto kV 80 - 120 kVp	37.4 ± 21.2 kg No BMI provided

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
Wang et al. 2021 [64]	A preliminary evaluation of applying DLIR in low-kilovolt scanning of upper abdomen	Prospective	Adult	Staging/ restaging	64	CECT	Liver Spleen Adipose tissue Paravertebral muscle Renal cortex	Low Medium High	ASIR-V 40% and 80% FBP	Group A, arterial phase mean = 10.8 mGy $\pm$ 0.9, portal phase mean = 5.5 mGy $\pm$ 0.4 Group B both arterial phase and portal phase mean = 13.6 mGy $\pm$ 1.5	80,100, 120	Group A 22.3 $\pm$ 2.9 kg/m ² Group B 23.4 $\pm$ 3.2 kg/m ²
Wang et al. 2022 [65]	The value of using DLIR for thinner slices to balance the image noise and spatial resolution in low-dose abdominal CT	Retrospective	Adult	Low dose chest CT indication	89	NCCT	Liver Subcutaneous Fat Paraspinal muscle Aorta	Low Medium High	ASIR-V 40%	Mean = 2.9 mGy $\pm$ 1.1	120	25.4 $\pm$ 4.8 kg/m ²
Xie et al. 2021 [66]	Application of DLIR under different radiation doses: focus on noise, contrast-to-noise ratio and image quality	Prospective	Adult	Liver lesions	17	CECT	Abdominal aorta Portal vein Liver Spleen Paravertebral muscle	Low Medium High	ASIR-V 30% and 70%	10.3 mGy, 2.6 mGy, 1.7 mGy	80	23.5 $\pm$ 2.5 kg/m ²
Xu et al. 2022 [42]	Quantitative and qualitative assessments of DLIR in low-keV virtual monoenergetic DECT	Retrospective	Adult	Not specified	30	CE - DECT	Liver Paraspinal muscle Aorta	High	ASIR-V 60%	Mean = 12.3 mGy $\pm$ 3.2	DECT 80 140	25.3 $\pm$ 5.1 kg/m ²
Yang et al. 2021 [35]	Impact of novel DLIR on diagnosis of contrast-	Retrospective	Adult	Suspected liver tumors	42	CECT	Liver	Low Medium High	ASIR-V 70% and 30%	Not reported	120	Not provided

Study	Abbreviated Title	Study type	Patient Age	Patient Health Status	Number of Patients	Scan Type	Anatomy Evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
	enhanced liver computed tomography imaging; Comparing to ASIR-V											
Yoo et al. 2022 [67]	Evaluation of Abdominal CT Obtained Using DLIR Compared with ASIR-V	Retrospective	Adult	Not specified	120	CECT	Liver	Low Medium High	ASIR-V 30%, 50%, 100% FBP	Mean = 5.1 mGy ± 1.9	100	23.1 ± 3.6 kg/m ²
Yoon et al. 2021 [68]	Image quality assessment of pediatric chest and abdomen CT by DLIR and ASIR-V	Retrospective	Pediatric	Not specified	23	CECT	Lungs Paraspinal muscles Liver Aorta	Medium High	ASIR-V 50% and 100%	Mean = 1.5 mGy ± 0.6	70-100	20.3 ± 5.1 kg/m ²
Zhang et al. 2022 [69]	DLIR in pediatric abdominal and chest computed tomography: a comparison of image quality and radiation dose	Prospective	Pediatric	Not specified	20	NCCT	Liver Kidney Subcutaneous fat Lung Spine Mediastinum	Medium High	ASIR-V 30% and 70%	Mean = 1.8 mGy ± 0.1	80	Not provided

CECT – contrast enhanced CT; NCCT – non-contrast CT; SECT – Single energy CT; DECT – dual energy CT; ASIR - Adaptive statistical iterative reconstruction; SNR - Signal-to-noise-ratio; CNR - Contrast-to-noise-ratio; DLIR - Deep learning image reconstruction; DLR - Deep learning reconstruction; NPS- noise power spectrum; MBIR - Model-based iterative reconstruction; SD - Standard dose; RD – Reduced dose; LD – Low dose

Characteristics for studies assessing Advanced Intelligent Clear-IQ Engine (AiCE) in abdominal CT

Table 2.

Study	Abbreviated Title	Study type	Patients	Health Status	Number of patients	Imaging type	Anatomy evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
Akagi et al. 2019 [70]	DLR improves image quality of abdominal ultra-high-resolution CT	Retrospective	Adult	Liver staging	46	CECT	Live Aorta Portal Vein	Not provided	HIR MBIR	Median = 12.6 mGy (Range, 9.7–16.1 mGy)	120	Not provided
Akagi et al. 2020 [52]	DLR of equilibrium phase CT images in obese patients	Retrospective	Adult	Obese patients for liver staging	50	CECT	Liver	Not provided	HIR MBIR	Median = 10.3 mGy (Range, 6.3–18.6 mGy)	120	High BMI Group 27.4 kg/m ² (Range, 25.2–32.9 kg/m ²) Low BMI group 21.6 kg/m ² (Range, 17.5–24.7 kg/m ²)
Brady et al. 2021 [18]	Improving Image Quality and Reducing Radiation Dose for Pediatric CT by Using DLR	Retrospective	Pediatric	Mixed indications, unspecified	19	CECT	Liver	Not provided	FBP SBIR MBIR	Control group mean = 2.1 mGy ± 1.1 DLR group mean = 1 mGy ± 0.7	Not specified	45 ± 26 kg (Range, 6–96 kg) BMI not provided
Nagayama et al. 2022 [28]	Radiation dose optimization potential of DLR for multiphase hepatic CT: A clinical and phantom study	Retrospective	Adult	Unspecified	73	CECT	Liver	HIR (Adaptive Iterative Dose Reduction 3D; AIDR-3D) and DLR (AiCE, BODY-SHARP) algorithms at a standard noise reduction level	HIR	Standard dose mean = 8.0 mGy ± 2.5, Low Dose mean = 16.8 mGy ± 3.4 *	120	23.9 ± 3.7 kg/m ²
Nakamura et al. 2019 [7]	DLR: Initial Evaluation Targeting Hypovascular	Retrospective	Adult	Hypovascular hepatic metastases	58	CECT	Liver	Not provided	HIR	Median = 10.3 mGy (Range, 5.0–23.8 mGy)	120	22.2 kg/m ² (Range, 14.7–33.5 kg/m ²)

Study	Abbreviated Title	Study type	Patients	Health Status	Number of patients	Imaging type	Anatomy evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
	Hepatic Metastases											
Nakamura et al. 2021 [29]	Diagnostic value of DLR for radiation dose reduction at ultra-high-resolution	Retrospective	Adult	Liver screening or staging	72	CECT	Liver Paraspinal muscle Aorta	1024 matrix, unknown DLR settings	HIR	Low dose median = 11.3 mGy (Range, 10.1–15.0 mGy) Standard dose median = 16.3 mGy (Range, 13.0–21.7 mGy)	Not specified	Lower Dose 22.2 kg/m ² (Range, 18.5–25.7 kg/m ²) Standard Dose 22.2 kg/m ² (Range, 16.8–27.5 kg/m ²)
Oostveen et al. 2022 [49]	Evaluation of Thin (0.5-mm) Slices Using DLR	Retrospective	Adult	Mixed, staging/restaging and acute abdomen	50	CECT	Liver	Body sharp	HIR	median = 3.0 mGy	100/120 (sure kV)	Not provided
Singh et al. 2020 [20]	Image Quality and Lesion Detection on DLR and IR of Submillisievert Chest and Abdominal CT	Prospective	Adult	Mixed, staging/restaging, pain, weight loss	59	CECT	Lungs Liver	Not provided	FBP HIR	SDCT mean = 13 mGy ± 4.4 LDCT mean 2.1 mGy ± 0.8	120	27 ± 5 kg/m ²
Tamura et al. 2021 [71]	Superior objective and subjective image quality of DLR for low-dose abdominal CT imaging in comparison with MBIR and FBP	Retrospective	Adult	Not specified	90	CECT	Liver	Body, mild	FBP MBIR (Body, mild)	Mean = 8.2 mGy (Range, 3.1–29.4 mGy)	120	25.5 kg/m ² (Range, 17.8–51.8 kg/m ²)
Tamura et al. 2022 [33]	DLR allows low-dose imaging while maintaining image quality: comparison of DLR and HIR	Retrospective	Adult	Suspected liver tumors (Hepatocellular carcinoma, N=34)	71	CECT	Liver	Not provided	HIR	AICE group mean = 10.8 mGy ± 3.9 AIDR-3D group mean	120	24.3 kg/m ² (Range, 17.8–32.2 kg/m ²)

Study	Abbreviated Title	Study type	Patients	Health Status	Number of patients	Imaging type	Anatomy evaluated	DLR Reconstruction Strength	Control Reconstruction Type	Radiation Dose (CTDI _{vol})	kVp	BMI/Weight
										≈ 19.4 mGy $\pm$ 5.6		
Zhang et al. 2022 [36]	Value of DLR at ultra-low-dose for urolithiasis	Prospective	Adult	Suspected urolithiasis	60	NCCT	Liver Kidney Spleen Uterus Prostate Aorta Psoas Muscle	Body sharp	HIR	LDCT mean = 6.8 mGy $\pm$ 1.0 ULDCT mean = 1.6 mGy $\pm$ 0.4	100	Not provided
Zhang et al. 2022 [37]	Application of DLR of ultra-low-dose in diagnosis of renal calculi	Prospective	Adult	Suspected renal calculi	60	NCCT	Liver Spleen Aorta Kidneys Psoas Major	Not reported	HIR	LDCT mean = 3.4 mGy $\pm$ 0.5 ULDCT mean = 0.80 mGy $\pm$ 0.19	100	Not provided

CECT – contrast enhanced CT; NCCT – non-contrast CT; SECT – Single energy CT; DECT – dual energy CT; AICE - Advanced intelligent Clear-IQ Engine; AIDR - Adaptive iterative dose reduction; SNR - Signal-to-noise-ratio; CNR - Contrast-to-noise-ratio; DLR - Deep learning reconstruction; NPS- noise power spectrum; MBIR - Model-based iterative reconstruction; SD - Standard dose; RD – Reduced dose; LD – Low dose; ULDCIT-Ultralow dose CT; HIR-Hybrid iterative reconstruction

Table 3.

Outcomes for studies assessing TrueFidelity in abdominal CT

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
Cao et al. 2020 [25]	Determining if DLIR can achieve improved image quality in contrast-enhanced abdominal CT with reduced radiation doses	Yes	CT number CNR Noise	Noise Sharpness Artifacts Diagnostic Confidence	DLIR-H significantly reduces image noise and generates images with clinically acceptable quality and diagnostic confidence with 76% dose reduction	DLIR High reduced noise by 40.8% compared to AV 50	Compared different contrast phases No observer performance assessed BMI not provided	12
Cao et al. 2022 [53]	Evaluation of how DLIR can improve image quality and diagnostic confidence of thin-slice contrast-enhanced abdominal CT	No	Edge Rise Slope Noise, CT Number CNR	Noise Contrast Small Structure Visibility Object Sharpness Lesion Contour	DLIR significantly reduces image noise and maintains a more natural image texture; image spatial resolution and diagnostic confidence can be improved using thin slice images and DLIR in abdominal CT	DLIR improves thin slices compared to AV50 (DLIR H slightly better than DLIR M); at 1.25 mm, DLIR High reduced noise by 34.8% compared to AV50	No observer performance assessed BMI not provided	12
Delabie et al. 2022 [48]	Assessing feasibility of DLIR on low-dose non-enhanced abdominopelvic CT images for detection of urolithiasis	No	CT Number Noise CNR SNR Absolute Contrast	Image Quality	For low-dose abdominopelvic CT, DLIR reconstruction exhibited image quality superior to ASIR-V and FBP as well as an excellent detection of urinary stones	DLIR inferior for < 3mm stones; Fig 3 shows small stone less dense on DLIR ("no clinical impact") 45% noise reduction DLIR H compared to AV70	Reference standard relies on the concomitant analysis of FBP and ASIR-V by the two readers	14
Fair et al. 2022 [40]	Evaluating DLIR on oncologic and vascular pelvic, chest and abdomen CT in obese patients	No	CT Number Noise CNR SNR	Texture Noise Sharpness Artifact Overall Image Quality	TrueFidelity DL image reconstruction provides improved image quality compared with ASIR-V in dual-energy CTs obtained in obese patients	Noise unspecified; DLIR SNR was 32% higher than AV	Noise not directly reported Lower than expected interreader agreement No observer performance assessed	10
Fukutomi et al. 2022 [39]	To compare DLIR and other reconstruction methods in terms of their effect on the accuracy of	No	Noise	N/A	DLIR algorithm reduced image noise and variability of iodine concentration values compared with other	DLIR M reduced noise 22% compared to AV 50	Only measured iodine concentrations in normal anatomic	14

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
	iodine quantification and image quality of dual-energy CT				reconstruction algorithms in the fast-kilovoltage-switching dual-energy CT		regions No observer performance assessed	
Ichikawa et al. 2021 [54]	Evaluating the feasibility of improving image quality of abdominal CT using DLIR in comparison with IR	No	CT number SNR CNR Noise	Noise Overall Image Quality	Image noise, overall image quality, CNR and SNR for abdominal CT images are improved with DLIR compared to hybrid IR	DLIR High reduced noise 37.3 % to AV50	No observer performance assessed	11
Jensen et al. 2019 [6]	Quantitatively and qualitatively evaluating DLIR in contrast-enhanced oncologic CT of the abdomen	No	CT number Noise CNR	Overall Image Quality Artifacts Lesion Diagnostic Confidence	Compared with 30% ASIR-V, DLIR improved CT evaluation of the abdomen in the portal venous phase. DLIR strength should be chosen to balance the degree of desired denoising for a clinical task relative to mild blurring, which increases with progressively higher DLIR strengths.	DLIR-H reduced image noise by 47% in comparison with ASIR-V 30%.	Single radiation dose level per patient No observer performance assessed	12
Jensen et al. 2022 [19]	Comparing image quality and liver metastases between reduced dose DLIR and standard dose FBP	Yes	CT number Noise SNR CNR Accuracy Sensitivity Specificity	Artifacts Image Texture Qualitative resolution	Deep learning image reconstruction (DLIR) improved CT image quality at 65% radiation dose reduction while preserving detection of liver lesions larger than 0.5 cm. Reduced-dose DLIR demonstrated overall inferior characterization of liver lesions and reader confidence.	DLIR Med 4.5 % reduced noise to AV60	Study assessed only two dose levels	13
Kaga et al.2021 [55]	Evaluating the feasibility of DLIR on contrast-enhanced abdominal CT and compare its performance across different strength	No	CT number Noise CNR SNR	Image Quality Diagnostic Acceptability	DLIR improved the SNR, CNR, and image quality compared with 40% ASIR-V, while making it possible to decrease lesion conspicuity using higher reconstruction strength.	DLIR H reduced noise 46.4% to AV40 (Supp. Data)	No observer performance assessed BMI not provided	11
Kaga et al.2022 [26]	Assessing the feasibility of DLIR in unenhanced low-dose abdominal CT	Yes	CT number Noise SNR	Overall Image Quality	DLIR facilitates substantial radiation dose reduction of > 75% and significantly reduces background noise. DLIR can maintain image quality and	LDCT DLIR High reduced noise 15% to SDCT AV40	No lesions	12

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
					anatomical structure depiction in unenhanced abdominal LDCt.			
Li et al. 2021 [27]	To investigate the feasibility of deep learning reconstruction of low-dose CT in achieving diagnostic images	Yes	CT number CNR Noise	Image Quality Lesion Diagnostic Confidence	DLIR algorithm improves image quality under low-dose scan condition and may be used to reduce the radiation dose without adversely affecting the image quality.	DLIR H was best; qualitatively improved small lesions; DLIR High reduced noise by 44.4% relative to AV40	Early arterial phase was compared to low dose late arteficial phase	13
Lyu et al. 2021 [34]	To examine the effect of DLIR in diagnostic performance and reader confidence in predicting resection of pancreatic cancer at CT	No	Noise Index Noise Power Spectrum Modulation Transfer Function Sensitivity Specificity AUC	Image Noise Sharpness Lesion Consistency Image Quality	The DLIR-H algorithm may improve the diagnostic performance and reader confidence in the CT assignment of the local resectability of pancreatic cancer while reducing the interreader variability.	DLIR High reduced noise by 40.1% compared to AV60	Surgical reference standards for local resectability were not available for some patients	13
Lyu et al. 2022 [21]	Comparing DLIR with IR and FBP in image quality and hepatic metastasis detection	Yes	CNR Detection Rate	Noise Distortion Sharpness Overall Image Quality	DLIR enables a 50% dose reduction for detecting low-contrast hepatic metastases while maintaining comparable image quality to full-dose FBP and IR.	Noise not specifically reported; Radiation reduction of 70% by DLIR performed inferiorly in detecting small metastases (< 1 cm) compared to full-dose FBP	The control/standard dose may be too low	14
Nam et al. 2021 [56]	Assessing the image quality of chest CT focusing upper abdomen CT image reconstructed using commercial DLIR in comparison with abdominal CT reconstructed by ASIR-V	Yes	CT number SNR CNR Noise	Noise Overall Image Quality Spatial Resolution Artifact	With < 50% of the radiation dose, DLIR chest CT showed comparable image quality in the upper abdomen to that of dedicated abdominal CT and was preferred by most readers.	RD DLIR High reduced noise 45.9% to RD AV50 and 34.9% to SD AV40	Compared different contrast phases BMI not provided	14

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
Njålstad et al. 2021 [57]	Assessing DLIR-reconstructed abdominal CT image quality and comparing them to images reconstructed with IR	No	CT number CNR Noise	Image Quality Noise Image Texture Artifact	Abdominal computed tomography images reconstructed using a DLIR technique shows improved image quality when compared to standardly applied iterative reconstruction across a variety of clinical image quality criteria.	Also assessed thin slices; DLIR high reduced noise 41.2% compared to AV50	Small number of patients No lesions BMI not provided	11
Noda et al. 2021 [30]	Evaluation of image quality and lesion detection in low-dose portal venous phase whole-body CT images reconstructed using DLIR	Yes	CT Number Noise SNR Lesion Detection Rate	Image Quality Sharpness Noise Diagnostic Acceptability	D CT images reconstructed with DLIR enable radiation dose reduction of >75% while maintaining image quality and lesion detection rate and superior SNR in comparison to SD-IR.	LD DLIR High reduced image noise by 19.7% to SD AV and by 43% to LD AV	Scans were at different timepoints Heterogeneous set of lesions	13
Noda et al. 2021 [58]	To compare DLIR and IR in terms of image noise, image quality and conspicuity of pancreatic ductal adenocarcinoma in pancreatic low-dose CT (LDCT)	No	CT Number CNR SNR	Image quality Conspicuity of Pancreatic Adenocarcinoma	DLIR significantly reduced image noise and improved image quality in pancreatic LDCT images compared with hybrid-IR.	DLIR High reduced noise by 54% compared to AV40	Low BMI patients	11
Noda et al. 2022 [59]	To compare the image quality and iodine concentration (IC) measurements in DLIR and IR on pancreatic protocol dual-energy computed tomography (DECT) images.	No	SNR CNR Background Noise CT number	Noise Sharpness Diagnostic Acceptability Conspicuity	DLIR could significantly improve image quality and reduce the variability of iodine concentration values than could hybrid-IR.	DLIR High reduced noise by 44.8% compared to AV40	No observer performance assessed	13
Noda et al. 2022 [31]	To assess if DLIR can effectively reconstruct thoraco-abdomino-pelvic CT with simultaneous reduction in radiation dose and iodine dose	Yes	Background Noise CT number SNR	Noise, Sharpness Artifact Diagnostic Acceptability	Virtual monochromatic images at 40 keV reconstructed with DLIR could achieve half doses of radiation and iodine while maintaining diagnostic image quality.	At RD 40 keV compared to SD 120 kVp, DLIR High was 71.4% higher noise but resulted in matched SNR	Low BMI patients No observer performance assessed	14

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
Parakh et al. 2021 [60]	Comparing images produced by DLIR reconstruction with ASIR-V in single energy CT images	No	CT Number Noise CNR	Noise Contrast Artifact Small Structure Visibility Sharpness	Sinogram-based deep learning image reconstructions were preferred over iterative reconstruction subjectively and objectively due to improved image quality and lower noise, even in large patients. Use in clinical routine may allow for radiation dose reduction.	DLIR-H reduced image noise by 48.7% compared to AV40	No observer performance assessed BMI not provided	12
Park et al. 2020 [61]	Comparing lower extremity CT angiography images produced by DLIR and ASIR-V in terms of noise and sharpness of vessels, liver, and muscle	No	CT number SNR CNR Noise Sharpness	Quantum Mottle	DLIR-High was the most balanced image in terms of image noise and sharpness among the examined image combinations.	DLIR High reduced noise by 9% compared to AV80	No observer performance assessed	12
Park et al. 2022 [32]	Comparing low-dose CT images of abdomen and pelvic produced by DLIR with standard-dose CT in terms of image quality and lesion detectability.	Yes	CT number Noise CNR SNR Lesion Detection	Overall Image Quality Noise Sharpness Texture Lesion Conspicuity	Overall image quality and detectability of focal liver lesions is preserved in contrast-enhanced abdominopelvic LDCT obtained with DLIR-M relative to those in SDCT with h-IR.	LDCT was 35.1% lower rad dose than SDCT; LD DLIR High was 45% lower noise than LD AV40 and 30.1% lower than SD AV40	Scans were at different timepoints	16
Sato et al. 2022 [62]	To assess the effectiveness of DLIR compared with IR in improving image quality of dual-energy CT (DECT)	No	CT Number CNR SNR	Image Noise Vessel Conspicuity Overall Image Quality	DLIR improves vessel conspicuity, CNR, and lesion conspicuity of virtual monochromatic and iodine density images in abdominal contrast-enhanced DECT, compared to hybrid IR.	DLIR High reduced noise 34-38% compared to AV50	No observer performance assessed	11
Son et al. 2022 [63]	Comparing DLIR with FBP and iterative reconstruction for reconstruction pediatric abdominopelvic CT.	No	Noise Power Spectrum Edge Rise Distance	Noise Edge Definition Overall Image Quality Lesion Detectability and Conspicuity Artifacts	For pediatric abdominopelvic CT, the DLIR algorithm may provide improved noise characteristics and better spatial resolution than the hybrid IR algorithm.	DLIR High reduced noise by 26% compared to AV50	ERD was measured only at the muscle and fat interfaces No observer performance assessed BMI not provided	15

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
Wang et al. 2021 [64]	Examine applicability of DLIR on a low - kilovolt enhanced scan of the upper abdomen.	Yes	CT number SNR Noise	Lesion Display Sharpness Noise Level	For upper abdominal low-kilovolt enhanced scan data, DLIR-H yields a more satisfactory image quality than FBP and ASIR-V.	DLIR high reduced image noise by 50% compared to AV40	A portion of the comparison was between different kVp No observer performance assessed	15
Wang et al. 2022 [65]	Comparing DLIR and ASIR-V 40% in terms balancing image noise and spatial resolution	No	CT Attenuation, Noise SNR Edge Rising Slope	Noise Artifacts Edge Sharpness of Organs Visualization of Small Structures	1.25 mm DLIR images had significantly reduced noise with improved SNR and overall image quality compared to AV40 at both 1.25 mm and 5 mm	DLIR-H reduced image noise by 53.2% compared to AV40	No observer performance assessed	14
Xie et al. 2021 [66]	Comparing DLIR and ASIR-V in enhanced upper-abdominal CT	Yes	CT Number Noise CNR	Vascular Significance (Small Structure Visibility) Noise Texture Lesion Conspicuity Lesion Diagnostic Confidence	DLIR can significantly improve the image quality of enhanced abdominal CT under the same dose. By decreasing the radiation dose to 150mAs, DL-H showed comparable image quality to that of RD-ASIR-V 30%.	DLIR High reduced image noise by 27 ~45.8% compared to AV30 at 5mm and 41.3 - 49.2% at 0.625 mm	No observer performance assessed Small sample size	12
Xu et al. 2022 [42]	Comparing DLIR for dual-energy CT (DECT) with ASIR-V in low kiloelectron volt (keV).	Yes	CT Number Noise SNR CNR	Image Noise Sharpness Texture Overall Image Quality	DLIR significantly reduced image noise compared to ASIR-V. Qualitative assessment showed that DLIR significantly improved image quality particularly in thin sliced images. DLIR may facilitate 40 keV as a new standard for routine low keV reconstruction in contrast-enhanced abdominal DECT.	DLIR high reduced image noise by 26% compared to AV60	No observer performance assessed	12

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome Measures	Conclusion	Noise Reduction	limitations	Article Quality Assessment Score
Yang et al. 2021 [35]	To compare image quality and lesion detection rate in images reconstructed using DLIR vs ASIR-V.	No	CT attenuation Noise CNR SNR Lesion Detection	Image Noise Score Artifact Score Hepatic Vein-Clarity Score Relative Liver Parenchyma Index Lesion-Clarity Score Relative Liver Parenchyma	Compared with AV-30 and AV-70, DLIR leads to better image quality with equal lesion detection rate for liver CT imaging under routine dose.	DLIR High reduced image noise by 57.3% compared to AV30, DLIR did not improve lesion detection - authors suggest value is more in that of dose reduction	CTDI _{vol} not reported heterogeneous set of liver lesions BMI not provided	13
Yoo et al. 2022 [67]	Comparing images produced by DLIR vs those produced by ASIR-V.	No	CT Number Noise	Image Quality Noise Image Sharpness	DLIR showed improved image quality and decreased noise under a decreased radiation dose.	DLIR High reduced image noise by 38% compared to AV50; DLIR Med was favored by readers over DLIR-High	No observer performance assessed	15
Yoon et al. 2021 [68]	Objective and Subjective comparison of DLIR and IR on pediatric chest and abdomen CT images.	No	CT number CNR SNR Noise Power Spectrum	Image Quality, Noise Artifact	Compared with 50% ASIR-V, DLIR improved pediatric body CT images with significant noise reduction. However, artifacts were not improved by DLIR, regardless of strength.	DLIR High reduced image noise by 38.7% compared to AV50; motion and beam hardening not reduced by DLIR	No observer performance assessed Small sample size	13
Zhang et al. 2022 [69]	To compare image quality and radiation dose between DLIR and ASIR-V produced abdominal and chest CT images.	Yes	CT Number Noise CNR SNR	Image Quality	Application of DLIR could help to reduce radiation dose without sacrificing the image quality of pediatric CT scans. Further clinical validation is required. DLIR superior to AV at 57% dose reduction	DLIR High reduced noise by 41.4-44.2% compared to AV30	No observer performance assessed BMI not provided	14

CECT – contrast enhanced CT; SECT – Single energy CT; DECT – dual energy CT; NCCT – non-contrast CT; ASIR – Adaptive statistical iterative reconstruction; SNR - Signal-to-noise-ratio; CNR - Contrast-to-noise-ratio; DLIR - Deep learning image reconstruction; NPS- noise power spectrum; MBIR - Model-based iterative reconstruction; SD - Standard dose; RD – Reduced dose

Table 4.

Outcomes for studies assessing Advanced Intelligent Clear-IQ Engine in abdominal CT

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome measures	Conclusions	Noise reduction	limitations	Article Quality Assessment Score
Akagi et al. 2019 [70]	Examining clinical applicability of DLIR on abdominal ultra-high-resolution CT (U-HRCT) in comparison with Hybrid-IR and MBIR	No	CNR CT number	Conspicuity Image Quality	DLIR improves the quality of U-HRCT better than both MBIR and HIR	AiCE reduced image noise by 44.2% to AIDR by 37.4% to FIRST	No observer performance assessed BMI not provided	13
Akagi et al. 2020 [52]	Identifying the reconstruction method that preserves diagnostic value of obese patients abdominal equilibrium phase CT images by comparing values in obese vs. non-obese patients.	No	Noise CNR	Image Quality	DLR preserved the quality of abdominal CT images obtained in obese patients.	AiCE reduced image noise by 30.5% to AIDR by 29.4% to FIRST	No observer performance assessed	11
Brady et al. 2021 [18]	To evaluate DLR ability to reduce radiation dose and improve quality of pediatric CT.	Yes	Object detectability (phantom)	Edge Definition Quantum Noise Level Object Conspicuity Overall Image Quality	DLR improved image quality and dose reduction without sacrificing noise texture or spatial resolution. Average object detectability for a 3-mm DLR image improved by 51% compared with FBP, 18% compared with SBIR, and 11% compared with MBIR (Fig 3).	DLR also showed 52% (1 of 2.1) estimated dose reduction than SBIR; 30.6% noise reduction of AiCE to AIDR	No observer performance assessed Small sample size BMI not provided	10
Nagayama et al. 2022 [28]	Comparing DLR and hybrid-IR impacts on image quality and radiation dose optimization for multiphase hepatic CT.	Yes	CT Number Noise CNR	The Noise Magnitude Texture Artifact Sharpness Interface Smoothness The Overall Quality	DLR improved the subjective and objective image quality of multiphase hepatic CT compared with HIR, even at approximately half the radiation dose.	In clinical study, LD-DLR yielded lower image noise, higher CNR, and higher subjective scores for all evaluation criteria than STD (all, $p \leq 0.05$), despite having 52.8% lower SSDE (8.0 ± 2.5 mGy vs. 16.8 ± 3.4 mGy).	No observer performance assessed	14
Nakamura et al. 2019 [7]	Assessing the effect of DLIR on conspicuity of hypovascular hepatic	No	Noise CNR	Overall Image Quality	DLR improved quality of abdominal CT for	AiCE reduced noise by 33.3% compared to standard AIDR	No observer performance assessed	11

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome measures	Conclusions	Noise reduction	limitations	Article Quality Assessment Score
	metastases in abdominal CT.				evaluation of hypovascular hepatic metastases.			
Nakamura et al. 2021 [29]	Evaluation of DLIR, hybrid-IR and MBIR on low dose ultra-high-resolution CT (U-HRCT) in comparison with reference standard which is defined as hybrid-IR on standard dose U-HRCT.	Yes	Noise CNR	Overall Image Quality	DLR preserved the quality of abdominal U-HRCT images even when scanned with a reduced radiation dose.	AICE reduced noise by 49% compared to AIDR	No observer performance assessed	12
Oostveen et al. 2022 [49]	Comparing DLIR and Hybrid-IR in thin-slice and thick-slice abdominopelvic CT.	No	CT Number Noise	Noise Sharpness Artifacts Natural Appearance Overall Image Quality	Thin-section DLR improves subjective image quality and reduces image noise compared with HIR, without causing artifacts.	0.5 mm AICE reduced noise 46% compared to 0.5 mm AIDR and 23% to 3mm AIDR	No observer performance assessed BMI not provided	13
Singh et al. 2020 [20]	Comparing image quality and lesion detectability in submillisievert chest and abdominopelvic CT images produced by DLIR vs those produced by IR.	Yes	CT Number Noise SNR Lesion Detection	Overall Image Quality Noise Texture	At submillisievert doses, DLR enables image quality and lesion detection superior to IR and FBP.	LDCT AICE was equivalent in noise to LD FIRST and it reduced noise by 26.7% to AIDR and 86.3% to FBP	LDCT was performed 5-10 seconds later than SDCT Mixed type of lesions included cholelithiasis	14
Tamura et al. 2021 [71]	Comparing image quality produced by DLIR with MBIR and FBP in abdominal CT.	No	CT number Noise CNR	Image Noise Artifacts Sharpness Overall Image Quality	Image quality of DLR was superior to MBIR and FBP on LDCT. DLR was only method with lower noise for obese patients than normal BMI.	AICE reduced image noise by 56.7% to FBP and 32% to FIRST	No observer performance assessed	15
Tamura et al. 2022 [33]	To compare radiation dose and image quality between DLIR and hybrid-IR	Yes	Noise CNR	Image Quality Diagnostics Acceptability	AICE was superior to AIDR for diagnostic acceptability. The use of AICE can reduce radiation dose by more than 40% without loss of quality compared to routine-dose abdominal CT with AIDR.	RD AICE reduced image noise by 6.1 % compared to SD AIDR	No observer performance assessed Scans were at different timepoints	14

Last name of the first author + Year	Aim	Radiation Dose Reduction Reported	Quantitative Outcome Measures	Qualitative Outcome measures	Conclusions	Noise reduction	limitations	Article Quality Assessment Score
Zhang et al. 2022 [36]	Comparing image quality and diagnostic accuracy in screening of urolithiasis between ULDCCT DLR and ULDCCT HIR while having LDCT HIR as reference	Yes	CT number SNR Detection Rate	Image Quality	ULDCCT-DLR performed comparably to LDCT-HIR in urinary stone detection and size estimation with better image quality and decreased radiation exposure.	A tiny stone became apparent on DLR compared to AIDR (contrary to other report showing two cases disappearing); DLR reduced noise by 46.7% compared to AIDR at ULDCCT	Only patients with calculi on LDCT underwent ULDCCT BMI not provided	14
Zhang et al. 2022 [37]	Evaluating effect of DLR on image quality, radiation dose, diagnostic capability and detection rate of renal calculi in ULDCCT	Yes	CT Number Noise CNR SNR Detection rate	Image Quality	At 76.6% dose reduction, DLR maintained image quality and renal calculi detection.	DLR reduced noise by 48.2% compared to AIDR at ULDCCT	DLR settings and BMI not provided	14

CECT – contrast enhanced CT; SECT – Single energy CT; DECT – dual energy CT; NCCT – non-contrast CT; AICE = Advanced intelligent Clear-IQ Engine; AIDR = Adaptive iterative dose reduction; SNR = Signal-to-noise-ratio; CNR = Contrast-to-noise-ratio; DLR = Deep learning reconstruction; NPS= noise power spectrum; MBIR = Model-based iterative reconstruction; SD = Standard dose; LD =low dose; RD=reduced dose; ULDCCT-Ultralow dose CT